ALWAYS ONE DIRECTION

Logarithmic functions are strictly monotonic: either always increasing or always decreasing throughout their domain. The base decides which.

MONOTONICITY

$f(x) = \log_a(x)$

split by **the base a**

CASE 1

$a > 1$

Strictly increasing

larger inputs produce larger outputs

$$x_1 < x_2 \Rightarrow \log_a(x_1) < \log_a(x_2)$$

E.G. $\log_2(x)$, $\log_{10}(x)$, $\ln(x)$

CASE 2

$0 < a < 1$

Strictly decreasing

larger inputs produce smaller outputs

$$x_1 < x_2 \Rightarrow \log_a(x_1) > \log_a(x_2)$$

E.G. $\log_{1/2}(x)$

TAKEAWAY

This is why taking logarithms preserves inequality direction when $a > 1$ and reverses it when $0 < a < 1$; see inequalities.